

Gender, language, and knowledge representation on Wikipedia

Final project

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Introduction

Wikipedia is one of the most widely used reference platforms in the world and plays a central role in shaping how cultural figures and historical knowledge are represented online. While Wikipedia presents itself as an open and neutral encyclopaedia, a growing body of scholarship has shown that its content reflects persistent inequalities related to gender, language, and cultural visibility. Biographical articles are a particularly important site for examining these dynamics, as they function both as records of individual achievement and as mechanisms through which cultural recognition is granted.

Previous research has demonstrated that women are underrepresented in Wikipedia biographies and that their articles often differ structurally from those of men, both in length and in the types of information emphasized (e.g. Reagle and Rhue (2011), Graells-Garrido et al. (2015)). At the same time, studies of multilingual Wikipedia have shown that language editions are shaped by distinct cultural and editorial communities, leading to systematic differences in how individuals are represented across languages (e.g. Callahan and Herring's (2011)). Together, this literature suggests that gender bias on Wikipedia cannot be understood independently of language and global visibility.

Building on these insights, the present study examines Wikipedia biographies of twentieth-century writers from Chile, Greece, and the Netherlands. By comparing men's and women's biographies within the same national contexts and across native-language and English Wikipedia editions, the project investigates how gendered patterns of representation interact with linguistic and cultural translation.

The central research question guiding this study is: *Do Wikipedia biographies of men and women differ systematically in their structure and level of detail, and do these differences vary between native-language and English editions?* To address this question, the analysis focuses on structural features of Wikipedia articles, which previous research has identified as indicators of editorial attention and perceived importance.

Based on prior scholarship, two hypotheses are proposed. First, *men's biographies are expected to be longer and more richly documented than women's biographies*, reflecting established patterns of gender bias in Wikipedia content (Graells-Garrido et al., 2015). Second, the study does not assume that English Wikipedia necessarily provides more extensive coverage than native-language editions. Instead, *it is hypothesized that English-language biographies may either exceed or fall short of native-language biographies depending on the individual's level of international recognition* (Callahan & Herring, 2011).

Literature review

Gender bias in Wikipedia biographies

Research on Wikipedia's gender gap has consistently demonstrated that men and women are not represented equally in biographical content. One early large-scale empirical study, by Reagle and Rhue (2011), compares biographical coverage in English Wikipedia and the Encyclopaedia Britannica. While Wikipedia contains more biographies of women in absolute numbers, the authors find that women's biographies are more likely to be missing relative to men's when compared to external reference sources. This finding highlights that gender bias persists even in open, collaborative knowledge environments and establishes biographies as a key site for studying inequality.

Importantly, gender bias is not limited to coverage alone. Graells-Garrido et al. (2015) show that men's and women's biographies differ systematically in structure and language. Their analysis reveals that women's biographies are more likely to emphasize personal relationships and private life, whereas men's biographies foreground professional achievements and intellectual contributions. These differences suggest that Wikipedia reproduces broader societal gender norms through its biographical narratives.

Together, these studies justify a direct comparison between men's and women's biographies, as adopted in the present project, which examines structural differences such as article length, references, and images.

Systemic gender inequality

While content-focused studies reveal how gender bias manifests in Wikipedia articles, other scholars address why such biases persist. Ford and Wajcman (2017) argue that Wikipedia's gender gap cannot be explained solely by the underrepresentation of women among editors. Drawing on feminist science and technology studies, they show that Wikipedia's infrastructure (including its norms of expertise, editorial policies, and technical systems) privileges historically masculine forms of knowledge production. These structural conditions shape not only who participates, but also what kinds of knowledge are produced and valued.

This systemic interpretation is reinforced by Ferran-Ferrer et al. (2023) scoping review of Wikipedia gender-gap research. Reviewing studies published between 2007 and 2022, they identify recurring patterns of content gaps, participation bias, and unequal visibility of women across all language editions. Crucially, they frame the gender gap as a systemic problem rather than an individual one, emphasizing that bias persists across languages and cultural contexts.

These insights support the present project's emphasis on differences in how biographies are structured and presented, rather than focusing solely on who contributes to Wikipedia, and they justify analysing men's and women's biographies as outcomes of broader sociotechnical inequalities.

Cross-language and cultural asymmetries

Wikipedia is not a single encyclopaedia but a network of language editions shaped by distinct cultural and historical contexts. Callahan and Herring's (2011) comparative study of English and Polish Wikipedia articles on famous persons demonstrates that language editions differ significantly in both the amount and framing of information they provide. Their findings point to a persistent English-language advantage, alongside culturally specific emphases in national editions. Importantly, English Wikipedia's centrality does not guarantee more comprehensive coverage for all subjects; research suggests that individuals with primarily national or regional significance may receive less detailed treatment in English-language editions than in their native Wikipedias (Callahan & Herring, 2011).

These differences have important implications for biographical representation. Readers relying on different language editions may encounter divergent portrayals of the same individual. For this reason, comparing native-language biographies with their English counterparts is essential for understanding how global visibility and cultural translation affect representation. The present project builds on this insight by first comparing men's and women's biographies within each language, and then examining how these gendered patterns shift between native-language and English Wikipedia editions.

Visual and citation bias across language editions

Representation on Wikipedia extends beyond textual content to include images and references, both of which shape reader perception and claims to credibility. He et al. (2018) demonstrate that visual content varies widely across language editions, even for articles about the same subject. Despite the shared infrastructure of Wikimedia Commons, most images appear in only one language edition, indicating that visual representation is culturally situated and editorially selective.

Citation practices reveal similar inequalities. Zheng et al. (2023) show that Wikipedia disproportionately cites scholarly publications authored by men and by researchers affiliated with Anglophone countries. Because references underpin Wikipedia's authority, such biases affect whose knowledge is legitimized. These findings provide a strong rationale for examining image counts and reference counts when comparing men's and women's biographies across languages.

Measuring article quality through structural features

Due to Wikipedia's scale, researchers often rely on structural features as proxies for article quality. Lewoniewski et al. (2016) demonstrate that indicators such as article length, reference count, and internal structure correlate with community-assessed quality across multiple language editions. Dang (2021) similarly emphasizes the central role of content-based features in computational quality assessment models.

Rather than treating these metrics as objective measures of accuracy, scholars interpret them as indicators of editorial attention and perceived importance. This approach aligns with the present project's use of structural features to compare gendered patterns of representation and to examine how these patterns vary between native-language and English Wikipedia editions.

The present research

Building on the preceding literature, this study addresses the intersection of gender and language in Wikipedia biographies by examining how structural inequalities manifest across different linguistic and cultural contexts. Existing research has demonstrated persistent gendered differences in biographical representation, as well as systematic variation between language editions, yet fewer studies explicitly combine these perspectives within comparable national and professional domains.

Together, prior scholarship highlights the importance of examining whether Wikipedia biographies of men and women differ systematically in their structure and level of detail, and whether such differences vary between native-language and English editions. Research consistently shows that men's biographies tend to be longer and more richly documented than women's, reflecting established patterns of gender bias in Wikipedia content (Graells-Garrido et al., 2015). At the same time, studies of multilingual Wikipedia suggest that English-language editions may privilege internationally recognizable figures over those with primarily national significance, resulting in uneven patterns of visibility across languages (Callahan & Herring, 2011).

By bringing these strands of research together, the present study situates gender bias within broader processes of linguistic and cultural translation on Wikipedia.

Data and methods

The dataset analysed in this study consists of eighteen writers from Chile, Greece, and the Netherlands, evenly divided by gender and active primarily in the twentieth century. Restricting the sample to a single professional field and historical period allows for meaningful gender comparison while reducing the influence of confounding factors related to occupation or era. The selection of three countries enables exploration of national differences while maintaining a manageable scope for close comparative analysis, and corresponds to the nationality of the participating students. The selected authors are the following:

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|----------------------|--------------------------|
| - Gabriela Mistral | - Pablo Neruda |
| - Isabel Allende | - Roberto Bolaño |
| - Marcela Paz | - José Donoso |
| - Annie M.G. Schmidt | - Harry Mulisch |
| - Hella Haasse | - Gerard Reve |
| - Renate Dorrestein | - Jan Wolkers |
| - Κική Δημουλά | - Ντίνος Χριστιανόπουλος |
| - Κατερίνα Γώγου | - Μανολής Αναγνωστάκης |
| - Αγγελική Πουκ | - Νίκος Καββαδίας |

Data were collected using the Wikipedia API and scraping techniques implemented in a Jupyter notebook. For each writer, both the native-language Wikipedia article and the corresponding English-language article were retrieved to ensure direct comparability. The analysis focuses on structural characteristics of the articles, including their length, number of references, and number of images. These features were selected because previous research has shown that they correlate with editorial investment and perceived importance within Wikipedia communities (Lewoniewski et al., 2016).

The analytical strategy combines gender comparison within each language edition with cross-language comparison of the same biographies. Ratios between English and native-language versions were calculated to examine how coverage changes when biographies move into English. Statistical tests and visualizations were used to identify patterns while maintaining a cautious interpretive stance, recognizing that quantitative differences do not directly translate into qualitative judgments about importance or value.

Results

The analysis demonstrates that gender differences in Wikipedia coverage persist across language editions, though these differences do not manifest in a uniform or linear way. Across the dataset, men's biographies are generally longer and more structurally developed than women's, a pattern that is particularly pronounced in English Wikipedia. This pattern is consistent with previously identified gender disparities in Wikipedia biographies. However, the data also reveal considerable variation within gender groups, especially among men, whose article lengths and documentation levels vary widely. This internal variation indicates that while gender remains a significant structuring factor, it does not operate in isolation. Some women's biographies, particularly in their native-language editions, are among the most extensive in the sample, demonstrating that high levels of documentation are possible for women within certain linguistic and cultural contexts.

A closer comparison between native-language and English Wikipedia articles complicates assumptions about English Wikipedia's comprehensiveness. Contrary to the expectation that English Wikipedia would consistently provide more extensive coverage, the analysis shows that English-language biographies are often substantially shorter than their native-language counterparts. This pattern is especially evident for writers whose recognition is primarily national or regional rather than international. Across all three countries examined, multiple cases show sharp reductions in word count, references, and images when moving from the native-language article to English. These reductions suggest that translation into English does not function as a neutral extension of visibility but often involves a process of selective condensation.

Gendered patterns become particularly visible when examining the ratios between English and native-language coverage. Women's English biographies tend to cluster well below parity, indicating a systematic loss of detail during the transition to English. Men's biographies, by contrast, exhibit much greater variability. While some men experience similar reductions, others show significant expansion in English, reflecting higher levels of international recognition or incorporation into global literary canons. This divergence suggests that the relationship between language and coverage is not symmetrical across genders. Instead, women are more consistently subject to reduction, while men's visibility is more contingent on their perceived global relevance.

The interaction between gender and language thus emerges as a central empirical finding. Women's biographies are more likely to shrink when translated into English, regardless of their prominence within national literary contexts. Men's biographies, on the other hand, are more likely to expand when their work circulates beyond national boundaries. This pattern points to gendered dynamics within processes of cultural translation and international canonization, where women's contributions appear less likely to be carried forward in full detail into global knowledge spaces.

Country-level analysis further underscores the importance of national context in shaping these dynamics. In the Chilean case, the contrast between men's and women's English-language coverage is particularly stark, with women experiencing more pronounced reductions relative to their native-language articles. The Netherlands shows substantial drops in English coverage for both genders, especially among writers who are highly canonical within national literary traditions but less internationally visible. Greece presents a somewhat different pattern, with generally lower levels of coverage across both language editions but less extreme divergence between men and women.

Visualizations of these results are provided in the figures below.

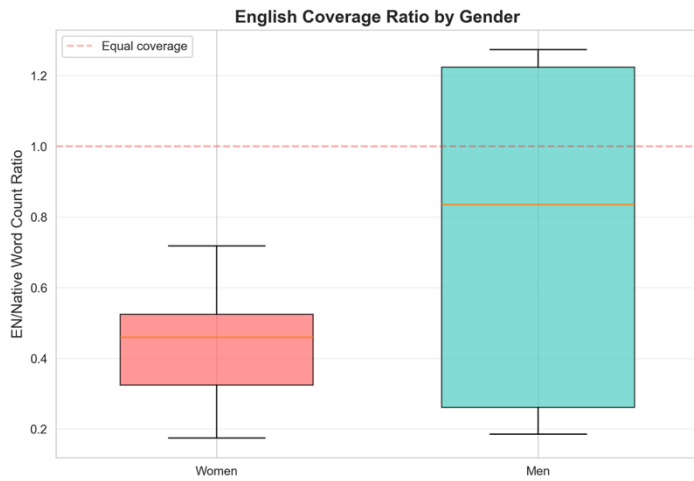


Figure 1
English Coverage Ratio by Gender

Note. Figure created by the author using Python (Jupyter Notebook).

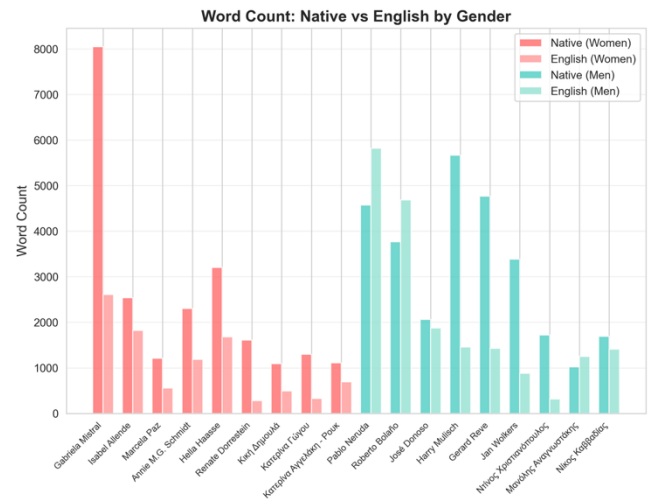


Figure 2
Word Count: Native vs English by Gender

Note. Figure created by the author using Python (Jupyter Notebook).

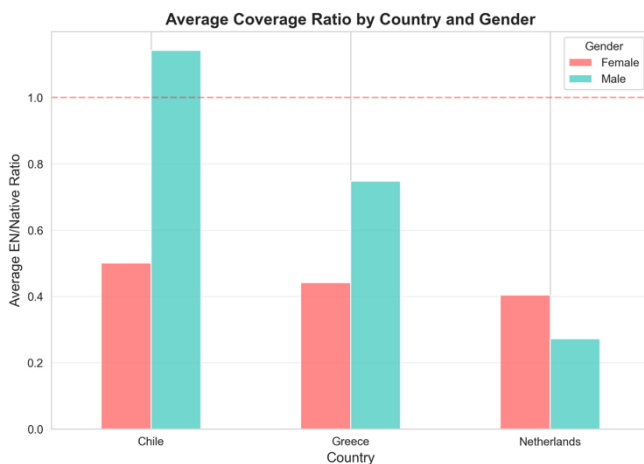


Figure 3
Average Coverage Ratio by Country and Gender

Note. Figure created by the author using Python (Jupyter Notebook).

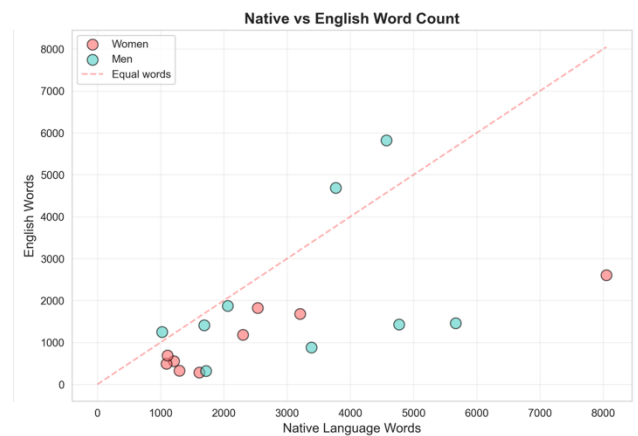


Figure 4
Native vs English Word Count

Note. Figure created by the author using Python (Jupyter Notebook).

Discussion

The following discussion interprets the results in relation to existing scholarship on gender bias in Wikipedia biographies, cross-language knowledge asymmetries, and sociotechnical systems of knowledge production. The findings demonstrate that gender and language interact in complex ways to shape the representation of notable individuals on Wikipedia, rather than producing uniform or predictable patterns of coverage.

Across the dataset, men's biographies tend to be longer and contain more references and images than women's biographies, particularly in the English-language edition. In relation to the first hypothesis, these results largely confirm earlier findings that women's biographies are structurally less developed than those of men, even when

women are clearly notable (Graells-Garrido et al., 2015; Reagle & Rhue, 2011). This suggests that gender bias on Wikipedia persists not only in terms of coverage, but also in the depth and richness of representation.

At the same time, the analysis shows that English Wikipedia does not consistently provide more extensive coverage than native-language editions. For many individuals, especially those whose cultural significance is primarily national or regional, English-language biographies are substantially shorter than their native counterparts. This finding supports the second hypothesis and aligns with Callahan and Herring's (2011) argument that English Wikipedia functions as a selective global filter, privileging figures with international visibility while offering more limited coverage of nationally canonical individuals. Importantly, this challenges the assumption that English Wikipedia can be treated as a universal benchmark for completeness or quality.

Gendered differences become particularly visible when examining how biographies change across languages. Women's biographies are more likely to lose content, references, and images when moving from native-language editions to English, whereas men's biographies show greater variability. This pattern suggests that international recognition interacts with gender, with women more likely to experience a reduction in visibility during processes of linguistic and cultural translation. Such findings resonate with prior research indicating that global knowledge platforms tend to reproduce existing hierarchies of gender and cultural authority.

The loss of images and references in women's English-language biographies is especially significant. Previous research has demonstrated that visual material and citation practices play a crucial role in establishing authority, credibility, and visibility on Wikipedia (He et al., 2018; Zheng et al., 2023). From this perspective, reduced visual and bibliographic presence does not merely reflect differences in article length, but points to deeper inequalities in how knowledge about women is legitimized and maintained across language editions.

Rather than attributing these patterns to individual editorial choices, the findings are best understood as the outcome of broader sociotechnical dynamics. Consistent with feminist critiques of Wikipedia, the observed disparities reflect how editorial norms, notability criteria, and infrastructural features jointly shape content production within unequal sociotechnical systems (Ford & Wajcman, 2017; Ferran-Ferrer et al., 2023). By focusing on representation outcomes rather than editor demographics, this study highlights how structural conditions influence whose knowledge is preserved and amplified.

Finally, the results underscore the importance of using structural indicators—such as word count, references, and images—as measures of editorial attention rather than objective quality. As emphasized by Lewoniewski et al. (2016) and Dang (2021), these metrics capture differences in investment and visibility, making them particularly suitable for comparative analyses of bias across language editions. While such indicators cannot account for narrative tone or implicit framing, they offer valuable insight into the uneven distribution of attention within collaborative knowledge systems..

Limitations

Several limitations on this study must be acknowledged. The relatively small sample size means that the findings should be interpreted as indicative rather than definitive. Differences in genre, historical positioning within the twentieth century, and levels of fame are not fully controlled. Structural metrics cannot capture narrative tone or qualitative framing, and observed correlations do not imply intentional bias. Moreover, Wikipedia is a dynamic platform, and this analysis represents a snapshot rather than a stable state.

Conclusion

This study demonstrates that gender bias in Wikipedia biographies cannot be understood independently of language and global visibility. Women's biographies are not only structurally thinner but are also more likely to be diminished when translated into English. English Wikipedia thus functions less as a neutral global archive and more as a gatekeeping space that selectively amplifies certain forms of cultural recognition. Visibility on Wikipedia is shaped by both language and fame, yet neither operates neutrally: cultural figures associated with languages that carry greater global significance tend to receive more international visibility, while fame itself is structured by gendered dynamics within a social system that rewards and amplifies men. For Digital Humanities research, these findings underscore the importance of multilingual, comparative approaches and of treating structural features as indicators of power rather than neutral measures of quality. Taken together, these findings demonstrate that gender bias on Wikipedia cannot be understood independently of language and cultural visibility, reinforcing the need for intersectional and multilingual approaches in digital humanities research on knowledge representation.

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